Automatic Streetlight Control System Using LDR

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***Abstract*—The increasing demand for energy efficiency in public infrastructure has motivated the development of intelligent street lighting systems. This paper details the architecture and simulation-based evaluation of an automatic street lighting system to reduce unnecessary electrical energy consumption while preserving road safety. The proposed system activates street lights only under required conditions , such as low ambient light or reduced visibility, and deactivates them when sufficient daylight is available. A microcontroller-based control unit processes input data from ambient light, motion, and humidity sensors to enable adaptive lighting operation. The system enhances illumination during poor visibility and conserves energy through dimmed lighting levels during low-traffic periods. Simulation results highlight the potential of the proposed strategy to improve power conservation while ensuring reliable and secure roadway operation.**

***Index Terms*—Automatic Street Light, LDR, Sensor.**

I. Introduction

Street lighting is a critical public infrastructure that supports nighttime mobility, enhances road safety, and contributes to public security in both urban and rural environments. In developing countries such as Sri Lanka, street lighting networks are widely deployed across large areas and are required to operate for long durations, resulting in significant energy consumption. As a result, the efficiency and reliability of street lighting systems have become important considerations due to increasing energy demand and sustainability concerns [1], [2].

Fig. 1. View of an illuminated street by automatic street light system

In Sri Lanka, different street lighting control systems are used instead of a single standardized method, depending on the area and available resources. In rural areas, street lights are still controlled manually, either through individual switching or group-based control systems.While most of urban and semi-urban regions mainly use automatic time-based control systems. Apart from these methods, street lighting in some selected areas is powered by grid-independent solar energy systems, thereby reducing reliance on the national electricity grid. This type of implementation largely depends on location, weather conditions, and the availability of maintenance resources.

Though traditional street lighting systems which are manual or fixed time based are commonly used because of their low cost and ease of maintenance , they are not suitable every conditions for like changing environmental system or traffic demand This can lead to unnecessary energy consumption.As a result streetlights may remain operational when illumination is not needed or provide uniform lighting without considering the traffic demand, leading to avoidable energy consumptionThese challenges are more severe in regions where both energy availability and maintenance resources are limited.

II. Literature Overview

Due to the above-mentioned reasons, researchers have begun exploring better solutions to reduce energy consumption while maintaining adequate visibility and public safety. Many automated and intelligent approaches have been proposed, incorporating sensing, adaptive control, and alternative power sources. The following section reviews existing technologies and methods related to street lighting systems

Street lighting systems have been widely studied due to their significant contribution to urban electricity consumption resulting from extensive deployment and long operating hours. Reports from global energy organizations and prior studies indicate that public lighting infrastructure represents a substantial portion of total urban energy usage, highlighting the need for improved efficiency in street lighting operation [1], [10].

Early street lighting control approaches, as discussed in the literature, primarily relied on manual operation and fixed time-based switching mechanisms. These approaches were shown to be energy inefficient because fixed schedules do not account for variations in daylight duration, seasonal changes, weather conditions, or actual road usage. Consequently, such systems may result in lights remaining operational during daylight

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hours or activating earlier than necessary, leading to avoidable energy wastage [1], [3].

To improve automation and reduce dependence on manual control, several studies have explored photocell- and Light Dependent Resistor (LDR)-based street lighting systems. These approaches regulate lighting operation by continuously monitoring ambient light intensity and activating lamps only when natural illumination falls below acceptable visibility thresholds. While such systems offer improved adaptability compared to time-based control, the literature reports reliability concerns associated with sensor aging, dust accumulation, calibration drift, and environmental sensitivity. These issues can lead to incorrect switching behavior, allowing streetlights to remain ON even under sufficient daylight conditions [4].

More recent research has investigated adaptive street lighting strategies that adjust illumination levels based on traffic density and human presence. Motion-based systems commonly employ Passive Infrared (PIR) sensors to detect vehicles or pedestrians and dynamically control light intensity. Studies report that such adaptive operation improves safety by providing full illumination when activity is detected while reducing energy consumption through dimming during low-traffic periods. Additional benefits identified in the literature include reduced light pollution and extended lighting component lifespan due to lower operational stress, making these systems particularly suitable for rural roads and low-traffic environments [5], [6].

Advances in smart city research have further introduced intelligent street lighting systems integrating microcontrollers, sensor networks, and IoT-based communication technologies. These systems enable capabilities such as individual lamp control, real-time monitoring, fault detection, centralized management, and predictive maintenance, resulting in improved energy efficiency and system reliability. However, multiple studies emphasize that high installation costs, increased system complexity, communication overhead, and maintenance requirements limit the feasibility of deploying such solutions in smaller cities and developing regions with constrained resources [2], [7].

Solar-powered street lighting systems have also been widely reviewed as an alternative to grid-dependent infrastructure. These systems utilize photovoltaic panels to generate electricity during the day and store energy in batteries for nighttime operation. Although long-term reductions in grid electricity usage are reported, existing studies highlight challenges such as high initial installation costs, battery maintenance and replacement requirements, and reduced reliability during extended cloudy or rainy conditions. As a result, solar-powered street lighting is often deployed selectively rather than as a universal solution [8], [9].

Another limitation identified in the existing literature is the limited use of simulation-based analysis prior to hardware implementation. Many studies emphasize physical prototyping without thoroughly evaluating sensor behavior, switching thresholds, hysteresis effects, or expected energy savings using simulation tools. The absence of such analysis restricts the ability to predict system performance under varying real-world conditions before deployment.

Therefore, despite the availability of various street lighting control strategies, a research gap remains in the development of a low-cost, reliable, and simulation-validated automatic street lighting system that reduces unnecessary energy consumption while maintaining safety and adaptability under diverse environmental and traffic conditions.

III. Problem Statement

Despite the widespread deployment of automated street lighting systems, the unnecessary operation of streetlights during daytime remains an observable problem. Streetlights are frequently found switched ON even when sufficient natural daylight is available, resulting in avoidable energy consumption, increased electricity costs, higher carbon emissions, and accelerated degradation of lighting equipment. This problem primarily arises from the continued use of basic and outdated control practices in street lighting systems. In many developing regions, lighting operation still relies on manual switching or fixed time-based control, both of which offer limited adaptability and depend on predefined schedules rather than actual environmental conditions. To reduce manual intervention, simple light-sensing mechanisms such as photocells have been introduced; however, these low-cost sensors are prone to malfunction due to aging, dust accumulation, improper calibration, and environmental exposure. Consequently, incorrect switching behavior may occur, allowing unnecessary energy consumption to persist.

Traditional street lighting solutions are mostly grid-dependent and maintain constant light intensity despite traffic demand , which is largely contributing to the electricity demand. Solar-powered streetlight technology has also been developed as an energy alternative with the intended objective of cutting the energy consumption load on the power grid and enhancing sustainability efforts. Although this energy technology reduces the dependence on the electric power grid, the environmental factors and the battery capacity limit its effectiveness as it may result in reduced light intensity or the entire system failing during the rainy season or cloudy days. However, grid-dependent lighting systems remain vulnerable to power outages caused by extreme weather or infrastructure failures, which could leave roads dark and endanger public safety, particularly in remote and rural areas. In addition, many existing lighting systems operate without considering real-time traffic demand. Continuous full illumination during periods of low or no traffic is common on rural and semi-rural roads resulting in unnecessary energy consumption regardless of the power source. Also in low-visibility conditions like fog or heavy rain extra illumination may be necessary even during the day to ensure the road safety

These difficulties point out the need for a street lighting system that can ensure reliable operation despite variations in power availability, minimize unnecessary energy consumption during times of low demand, and maintain safe illumination under changing environmental conditions. The design and simulation-based evaluation of an affordable and adaptable street lighting control system designed to improve energy efficiency while ensuring reliable and safe operation under varying environmental, traffic, and power supply conditions is the issue this study aims to address.

IV. Methodology

The following shows the flowchart for the process.

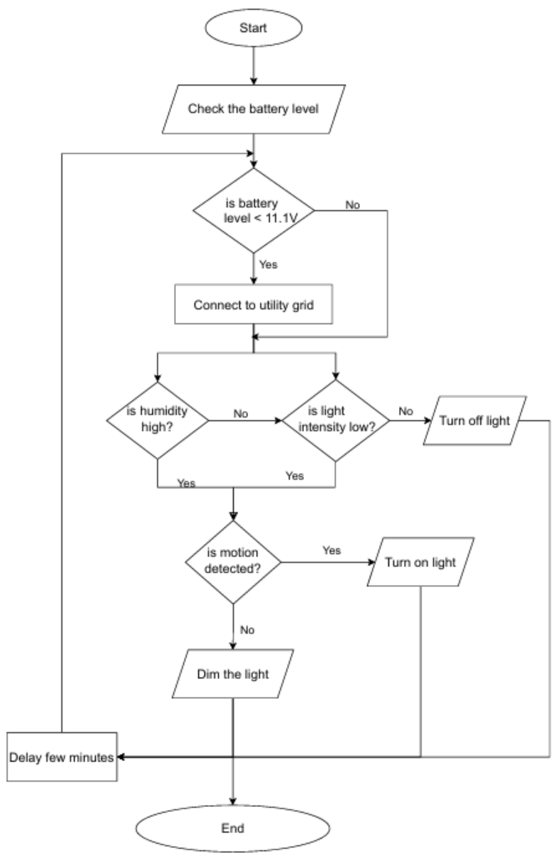

Fig. 2. Flowchart for process

A. Step 1: Problem Definition and Objective Setting

Problem: Conventional street lighting systems often rely on manual control, fixed schedules, or basic light sensors that are unable to adjust to changing daylight or weather conditions, which results in lights remaining on unnecessary and wasting energy. Because these systems rely on the electrical grid, roads may become dangerous and dark during power outages caused by extreme weather or infrastructure failures. They use more electricity than required because they typically operate at a constant brightness regardless of traffic demand, which raises costs and carbon emissions. Furthermore, outdated controls and simple sensors often malfunction or respond incorrectly, making it difficult for them to adapt to real-time conditions or low-visibility situations such as daytime fog

Primary Objectives: The primary objective of this work is to design an adaptive street lighting system that minimizes unnecessary energy consumption while maintaining road safety. This is achieved by:

- Adjusting lighting operation based on ambient light conditions and reduced-visibility scenarios such as fog
- Providing adaptive illumination using motion detection to avoid continuous full-brightness operation
- Prioritizing renewable solar energy while ensuring reliable operation through a hybrid grid-assisted power supply.

Key Constraints: The system design is constrained by limited solar energy availability and battery storage capacity, the need for reliable environmental sensing under outdoor conditions, and the requirement for uninterrupted operation during adverse weather or low-traffic periods. Additionally, the solution must remain cost-effective, simple to implement, and suitable for long-term deployment in exposed environments.

B. Step 2: Conceptual Design and Vision Development

The suggested system is conceptually built around an adaptive control approach that combines traffic-based lighting control, environmental sensing, and power management. To guarantee continuous operation, the system prioritizes solar energy by keeping monitoring on the battery level and automatically switching to the utility grid when the battery is low.. In order to determine visibility conditions for lighting control, humidity and ambient light intensity are evaluated in parallel. Low ambient light or high humidity indicates reduced visibility, which activates motion-based control that uses a Passive Infrared (PIR) sensor to identify the presence of vehicles or pedestrians. The street light operates at full brightness when motion is detected; otherwise, it stays dimmed to save energy. The light stays off when there is enough daylight and normal visibility conditions.The street light operates at full brightness when motion is detected; otherwise, it stays dimmed to save energy. The light stays off when there is enough daylight and normal visibility conditions. Physically, the system is designed as a steady pole-mounted , lighting unit with a curved support arm that securely positions the lamp over the road. All sensors, control electronics, and wiring are enclosed within the structure to protect them from environmental conditions.

Solar panels installed atop the pole serve as the main energy source, with a hybrid grid backup to maintain uninterrupted lighting. This design not only saves energy but also keeps roads safe under all traffic and weather conditions

C. Step 3: Detailed Design and Optimization

Mobility and Stability: Used a high-tensile metal pole and a reinforced concrete base to ensure the light stays stable in wind and rain.

Power System: Utilized solar panels to charge internal batteries, supporting the power demands of the sensors and the LED lamp throughout the night.

Navigation and Detection: Placed the LDR and DHT22 sensors near the top for clear air/light readings, and positioned the PIR sensor to detect movement on the ground accurately.

Durability: Routed all electrical cables internally through the hollow arm and pole to protect them from heat, moisture, and physical damage.

Lighting Capacity: Optimized the light output so that it only uses 100

D. *Step 4: Simulation*

Utilized circuit design layouts to organize the Arduino Uno, sensors, and LED connections to ensure a clean and functional electrical path.

Applied a detailed logic flowchart to simulate how the system reacts to different inputs (fog, darkness, and movement) to ensure the light responds correctly in every scenario.

Used 3D physical modeling to check the tilt of the solar panels and the reach of the curved arm for the best possible energy collection and light spread.

V. System Design Specifications

A. *Hardware Design*

The hardware layer consists of an Arduino Uno microcontroller acting as the central processing unit, interfaced with the following components:

- DHT22 Sensor: Monitors ambient humidity levels. High humidity or fog detection triggers higher visibility modes.
- LDR (Light Dependent Resistor) Module: Measures ambient light intensity to determine the transition between day and night cycles.
- PIR Motion Sensor (HC-SR501): Detects human or vehicle presence within a specific radius.(3m to 7m can be adjusted)
- LED Output: Represents the street lamp, controlled via Pulse Width Modulation (PWM) to allow for variable brightness (Dimming vs. Full Brightness).

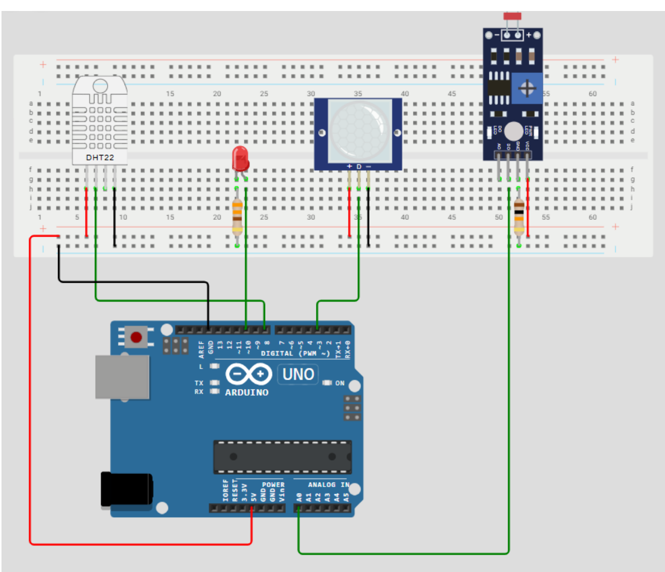

Fig. 3. System circuit arrangement

B. *Physical Construction*

1) Physical Design
The mechanical assembly of the street light consists of several functional parts integrated into a single pole-mounted unit:

- **Solar Panel Energy Array:** The top of the structure features two solar panels positioned at an angle to capture the most sunlight for battery charging.
- **Support Mast:** A strong cylindrical pole acts as the main frame for the system.
- **Stability Base:** The pole is attached to a heavy concrete base to keep the structure upright and stable.
- **Cantilever Support Arm:** A curved arm extends from the pole to hold the light over the road.
- **Internal Wiring:** All electrical wires connecting the solar panels, the controller, and the light are hidden inside the hollow pole and arm. This protects the cables from weather damage and keeps the design clean.

Component	Material Specification	Height/Length	Width / Diameter
Main Pole	Hot-dip Galvanized Steel (Q235/Q345)	6.0m – 8.0m	150mm (Base) to 80mm (Top)
Solar Mounting Bracket	Powder-coated Carbon Steel	1.0m – 1.5m	50mm – 60mm Tube
Light Arm (Curved)	Aluminum Alloy or Galvanized Steel	1.5m – 2.0m	60mm Diameter
Luminaire Housing	Die-cast Aluminum (IP66 Rated)	600mm – 800mm	250mm – 300mm
Battery/Control Box	SUS304 Stainless Steel (Grade 304)	400mm – 500mm	300mm x 200mm
Foundation Base	Reinforced Concrete (C25/C30 Grade)	1.0m (Underground)	600mm x 600mm
Anchor Bolts	High-tensile Steel (Grade 8.8)	500mm – 800mm	M20 – M24 Thread

Fig. 4. Physical structure measurements

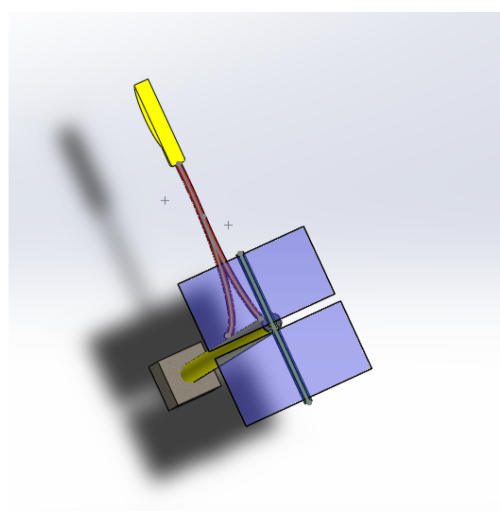

Fig. 5. Light pole top view

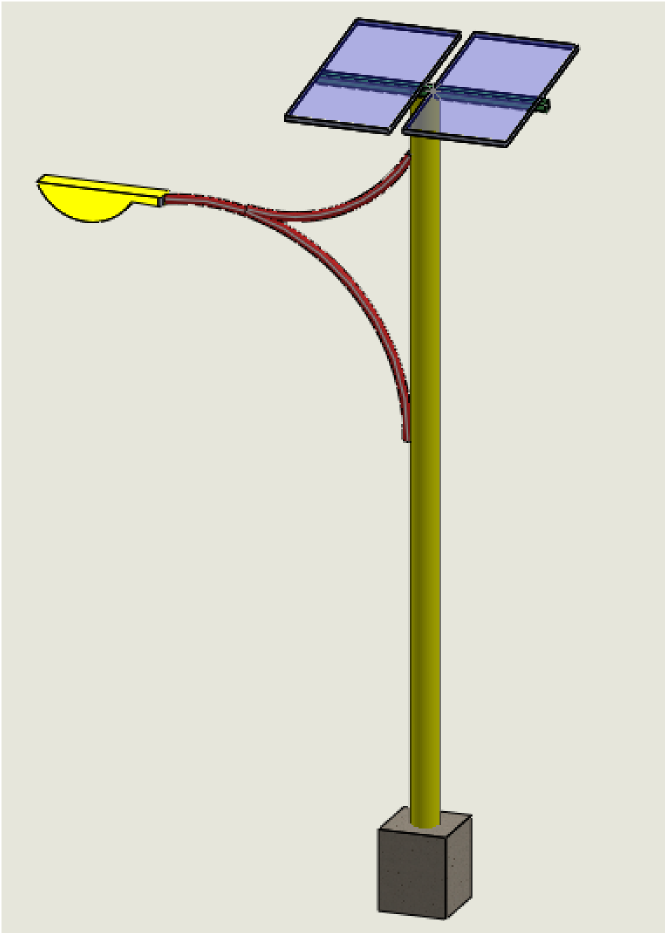

Fig. 6. Light pole front view

2) Control System Integration
The physical structure houses the sensors and the processing unit (Arduino) that manage the operation of the light:
   - **A. Sensor Placement** The sensors are placed strategically on the frame to gather accurate data:
     LDR and DHT22 Sensors: These are mounted near the solar panels to monitor ambient light levels and humidity (fog) without obstruction.
     PIR Motion Sensor: This sensor is placed on the lower part of the light or pole to detect the movement of people or vehicles below.

3) Power System Integration
The system is designed as a Hybrid power system, meaning that it relies on both solar power and the utility grid for its power requirement. In this setup, the solar panel and battery serve as the primary power source to maximise the use of renewable energy. The system switches to its secondary source, which is the utility system only when the battery has an inadequate amount of power. This can be used to overcome the main weakness of solar power which is its dependency on the weather. So by using this method we can ensure that the light works regardless of environmental conditions.

Note that the component values described below are representative of a practical system, designed for the real world implementation.

1) Solar panel and battery integration: Here, the battery powered by the dual solar panels helps to make sure that the system uses renewable energy and can remain functional independent of the utility grid. The following components are used in this circuit:

- **Solar Panels:** These dual solar panels are the primary energy generator for the power system. They capture sunlight and convert it to 12V DC current.
- **Battery (B1):** A Lithium Iron Phosphate (LiFePO4) battery with the capacity of 12V was used to store the energy generated by solar panels.
- **LM317T Voltage Regulator:** The safe charging threshold for the battery used in the system is 12V. So the voltage regulator takes the variable voltage from the solar panels and scales it down to a stable output to charge the battery safely.
- **Capacitor (C1):** A 1000µF capacitor is used to filter the electrical signal and remove noise and protect the regulator from sudden voltage spikes. It also helps to prevent surge during the switch from solar to grid and vice versa.
- **R1 (Program Resistor 240 ohm):** This resistor is used to maintain the 1.25V stable reference which is the industrial standard for the smooth operation of the voltage regulator.
- **R2 (Voltage Set Resistor 2.5kohm):** This combines with the R1 resistor and forms a voltage divider that helps to maintain the correct output during the charging process of the battery.

For the simulation model the values of these components have been scaled down to a 5V DC environment.

The values for the scaled down simulation ciruict are shown below.

Component	**Real-World System (Practical)**	**Simulation Model (Scaled)**
System Voltage	12V / 24V DC	5V DC
Solar Panel Capacity	100W – 200W	5W – 10W
Battery Capacity	60Ah – 120Ah (LiFePO4)	2,000mAh – 5,000mAh
Resistor R1 (Reference)	240Ω	220Ω
Resistor R2 (Voltage Set)	2.5kΩ	680Ω
Capacitor (Power Rail)	1000µF+ (Industrial)	100µF (Electrolytic)
Grid Integration	220V/110V AC Smart Switch	5V DC via USB Adapter
Area Covered	Full Roadway	Small Model / Desktop

Fig. 7. Practical and Simulation based values for the system

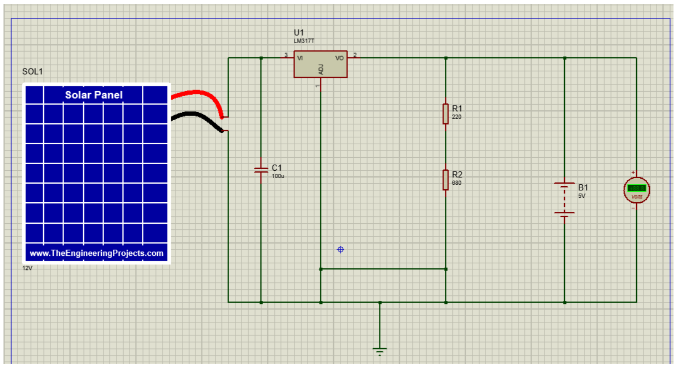

Fig. 8. Simulation circuit for the solar and battery power system

2) Grid Integration and battery monitoring system: This acts as a secondary power source for the system. The system makes the decision to switch over to the grid power based on monitoring two factors:

- **Battery level monitoring:** The system continuously monitors the battery level. If this level falls below a certain low threshold (11.5V – 12.0V), the system automatically switches the power source to the grid. This prevents the light from turning off and protects the battery from being completely drained which could affect the lifespan of the battery.
- **Smart Switching:** By looking at both the low battery level which indicates the remaining battery capacity the system can make the required choice. For example, if the battery is getting low he system switches to the grid immediately to ensure the street light stays on without interruption.

Feature	Real-World (12V Street Light)	Simulink Simulation (5V Model)
Nominal Battery Voltage	**12V**: Standard for lead-acid or LiFePO4 street light batteries.	**5V**: Used as a simplified logic level for the simulation model.
Switch to Grid (Low)	**11.1V** - **11.5V**: The "Low Voltage Disconnect" to protect battery life.	**3.5V**: The threshold programmed into Simulink Relay block.
Switch to Battery (High)	**12.6V** - **13.8V**: Requires the battery to reach a full or "float" charge state before reconnecting.	**4.0V+**: A mathematical buffer (hysteresis) to prevent the solver from crashing.

Fig. 9. Battery monitoring system parameter for a real-life system and simulation

Simulation input

- **Primary Power Source:** A *Simscape Battery Block* (Lithium-Ion) is the main energy storage unit.
- **Charging Unit:** A *Controlled Current Source* and the *constant block* connected to the battery's positive end, is used to represent the solar panels charging array.
- **Secondary Power Source:** A *DC Voltage Source* representing the 5V Utility Grid, which acts as a backup when the battery reaches its low threshold (in this case 3.5V) discharge level.
- **Diode:** is placed in series with the battery to prevent reverse current flow from the utility grid to the battery.

Switching Decision: When the battery voltage drops to 3.5V, the Relay outputs a logical 0 and the power source switches to utility grid.

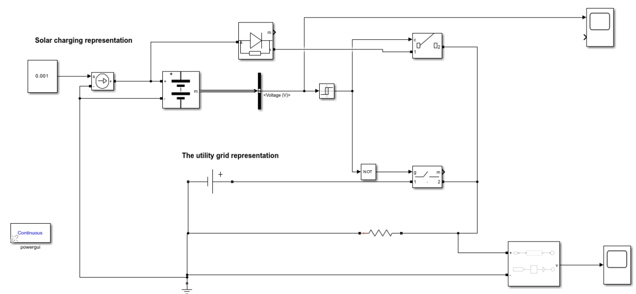

Fig. 10. Matlab simulink model for Battery monitoring system

Simulation output

The battery voltage level graph

As the battery powers the load (used to represent the LED bulb) the voltage level of the battery declines. After the battery voltage reaches the 3.5V at 890 seconds the battery disconnect from the load and it starts charging back up again thorough the controlled current source and the constant block.

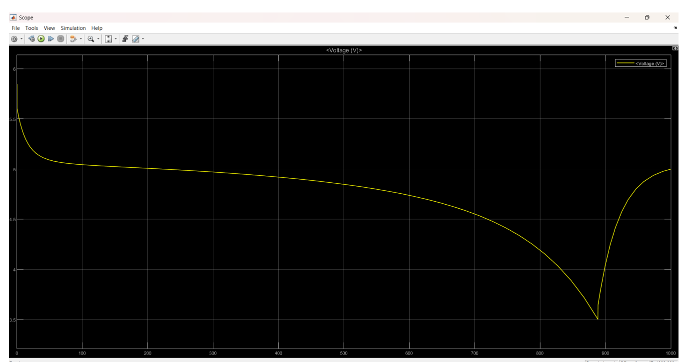

Fig. 11. The battery voltage level graph

The switching logic graph

This graph shows the source switching of the circuit. While the battery is discharging the graph also declines but when the battery voltage decreases to the 3.5V the signal goes back to 5V level. This is because when the battery hits 3.5V the power source switches to the utility grid continuing the unlimited power supply.

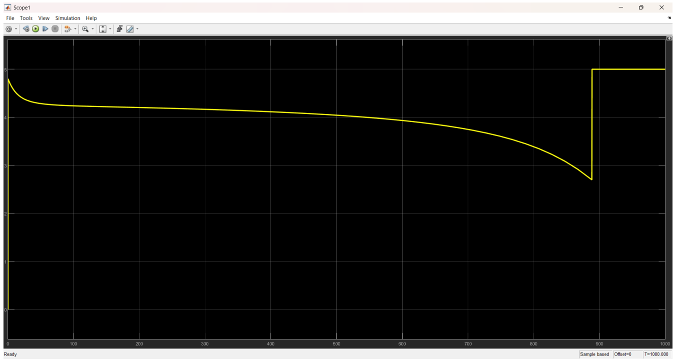

Fig. 12. The switching logic graph

4) Operational logic
   The light output is controlled by the logic defined in the system flowchart:
   - **Environmental Activation:** The system turns the light on if it detects high humidity (fog) or if ambient light levels are low (night).
   - **Adaptive Brightness:** To save energy collected by the solar panels, the light operates at full brightness only when motion is detected. When the area is empty, it switches to a dimmed state to reduce power use while keeping the area visible.

VI. Conclusion

A. Significance of product

By using a solar prioritized hybrid architecture, this of this automated street light system shows a functional and effective solution to minimize the energy wastage in the real world. The use of solar panels, which optimise sustainability of the system while keeping a utility grid connection as a safe backup power source, is one of this design's main advantages. The system's grid backup ensures steady function even in areas where solar power generating alone might not be enough. And by giving priority to solar generated power, it lowers operating costs and carbon footprint.

The system effectively combines this renewable energy source with a multi-sensor control logic to ensure reliable operation. The LDR and DHT22 sensors allow the light to respond to both nighttime and hazardous weather, such as heavy fog. Furthermore, the PIR motion sensor enables adaptive brightness, where the light provides full intensity only when movement is detected. This approach significantly preserves the energy collected by the solar panels, ensuring the battery can power the light throughout the night. Ultimately, this project proves that smart sensing and solar technology can work together to create a sustainable, safe, and independent lighting solution.

B. Suggestions for future research

To build upon this work, the following areas are recommended for future research:

- **IoT Integration:** Adding wireless modules (like the ESP8266) would allow each pole to send data to a central station. This would make it easier to track energy levels from the solar panels and identify if a lamp needs repair.
- **Networking Between Poles:** Research could be done on "daisy-chaining" the lights. If one pole detects motion, it could signal the next few poles to brighten in advance, creating a safer path for fast-moving vehicles.
- **Battery Technology:** Investigating different types of batteries, such as Lithium Iron Phosphate ($LiFePO_4$), could improve the system's ability to store power from the solar panels and increase the overall lifespan of the unit.
- **Automatic Maintenance:** Since dust can block the solar panels, future designs could include a self-cleaning mechanism or a special coating to ensure the panels always collect the maximum amount of sunlight.
- **Weather-Specific Algorithms:** The current logic uses humidity to detect fog. Future research could refine this by using different brightness levels for different levels of rain or snow to further save energy while keeping the road safe.

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